Shirpur Education Society's
R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR
An Autonomous Institute
(Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere)
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(स्वायत्त महाविद्यालय)

Programme: B.Tech in Artificial Intelligence and Machine Learning

Year: II/Semester III (Exam Year: 2024-2025)

Subject: Operating Systems **Time:** 09:00 am - 11:00 am (02:00 Hrs.)

Date: 20 Dec 2024 **Max Marks:** 60

END SEMESTER EXAMINATION ODD SEM-V (2024-2025)

Instructions:
1. This question paper contains 2 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Assume suitable data wherever required, but justify it.
5. Support your answers with neatly labelled diagrams, wherever necessary.

1. **15**

a. . **5**

i. Define OS. Explain system bus and its types in operating system. **5**

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ii. Describe architecture of a typical microkernel along with suitable diagram. **5**

b. Discuss the characteristics and applications of the following operating systems: **5**

i. Real-time Operating System

ii. Parallel Operating System

c. Explain deadlock and the four necessary conditions for deadlock to occur. **5**

2. **15**

a. 1. Explain each state of process state model with a neat diagram. **5**

b. . **10**

i. Explain FCFS CPU scheduling algorithm along with its advantages and disadvantages. **10**
Consider the following processes with burst time. Calculate the average waiting time and average turnaround time for FCFS and SJF (preemptive).

Process id	Arrival time	Burst time/CPU execution time
P1	0	2
P2	1	3
P3	2	5
P4	3	4
P5	4	6

---------- OR ----------

ii. Compare CPU scheduling algorithms with their merits and demerits along with example. **10**
Demonstrate priority scheduling with preemptive version with suitable example.

3. **15**

a. . **5**

i. Explain the various principles of concurrency. **5**

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ii. What is semaphore? Explain its types. **5**

b. Discuss producer consumer problem and provide the solution to the problem using monitor. **10**

4. **15**

a. Explain Banker's Resource Request Algorithm with simple example. **5**

b. . **10**

1. 1. Write short note on: **10**

I. Memory management requirements. [5 M]

II. Segmented allocation with suitable example. [5 M]

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2. 1. Discuss virtual memory management with paging. [5 M] **10**

Demonstrate FIFO and Optimal Page replacement policy. Page references: 6, 0, 1, 2, 0, 3, 6, 0, 3, 5, 6, 2, 4, 0 page frames : 3